

Final Review of the Project Status, Canonical Architecture, Remaining Blockers, and Required Corrections

This response has been reviewed against:

- the latest TSPI Project Status report;
- the corrected 39-Axis Master;
- the numbered 180 Living Biological Network Atlas;
- the final TSPI architecture;
- the latest Domain Expert confirmation concerning CELA.

The development team has made substantial and important progress. The software foundation is now strong, deterministic, registry-driven, and much safer than the previous report-generation system. However, the correct present status is:

The software core is substantially complete and technically tested. The canonical framework is available. Clinical mappings, evidence registries, Network edges, Module approvals, clinical validation, and production qualification are still incomplete.

The 71 passing automated tests demonstrate that the software behaves according to its implemented rules. They do not yet establish that the clinical knowledge, mappings, recommendations, or scoring algorithms are clinically valid. □

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1. Canonical TSPI architecture

The official architecture is:

3 Keys

→ 9 Restoration Steps

→ 12 Domains

→ 39 Biological Axes

→ 180 Living Biological Networks

→ 200 Phytochemical Modules

The latest Master identifies:

the 39 Axes as the stable biological assessment dimensions;

the 180 Networks as executable living biological systems;

the 200 Modules as the intervention layer. □

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Official 3 Keys

KEY-A — Metabolic Energy

KEY-B — Biological Dynamics

KEY-C — Biointegrity

The three Keys are concurrent biological dimensions. They are not sequential treatment stages.

The philosophical relationship between the Keys and traditional concepts may remain in explanatory documents, but it must not affect clinical scoring unless a separate canonical Key Philosophy Master is approved. The source Master contains some narrative variations in those philosophical correspondences; therefore, only the stable Key names and codes should enter the computational engine.

2. Important clarification: conceptual hierarchy is not the same as computational order

The Project Status currently describes the reasoning flow as:

Clinical Evidence

→ Evidence Integration

→ 3 Keys

→ 9 Restoration Steps

→ 12 Domains

→ 39 Axes

→ 180 Networks

→ Module Ranking

This is acceptable as a framework hierarchy, but it should not be implemented as a rigid sequential clinical inference pipeline.

The 3 Keys, 9 Steps, and 12 Domains serve different functions:

3 Keys are high-level biological interpretation dimensions.

12 Domains organize the 39 Axes.

9 Restoration Steps determine intervention sequencing.

39 Axes are the stable patient-assessment and Module-eligibility layer.

180 Networks are the dynamic mechanistic reasoning layer.

The recommended operational pipeline is:

Patient Data

→ Red-Flag and Critical-Value Screening

→ Data Normalisation and Provenance

→ Evidence Integration

→ Clinical Phenotype

→ Candidate Axis Assessment

→ Differential Network Model

→ Root-Driver and Compensatory-Network Analysis

→ Domain and Key Roll-Up

→ Restoration-Step Prioritisation

→ Module Eligibility

→ Safety Gate

→ Module Ranking and De-duplication

→ Physician Review

→ Reports

→ Clinical Outcome

→ Biological Model Update

The biological explanatory direction remains:

Living Network Dysregulation

- Symptoms
- Laboratory Changes
- Structural Changes
- Clinical Disease Expression

These two directions are not contradictory:

the first is the clinical inference workflow used when analysing patient data;

the second is the biological causal model used to explain how disease expression develops.

TSPI must not reason primarily from disease labels. Diseases remain useful clinical labels, but the biological reasoning units are the Living Biological Networks. □

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3. Official 39-Axis confirmation

The corrected 39-Axis list is the sole canonical Axis ontology.

Important confirmed Axes include:

- A8 — Mitochondrial Network
- A10 — Proteostasis–Autophagy–Lysosome
- A26 — Stem Cell & Hematopoiesis
- A27 — Glycation / Carbonyl Stress
- A34 — Neuro–Sleep–Stress–Brain Clearance
- A35 — Visceral Organ Functional Reserve
- A36 — Local Organ & Tissue Interface
- A37 — Oncologic Cell Fate
- A38 — Tumor Microenvironment & Metastatic Dynamics
- A39 — Genomic Regulation Homeostasis

The corrected Master explicitly states that proteostasis, UPR, autophagy, lysosome function, and protein aggregation belong under A10. A37–A39 are reserved for oncology and whole-system genomic regulation. □

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Any older definition that places:

A37 = Protein Quality Control

A38 = Protein Clearance

A39 = CELA

must be marked:

DEPRECATED

SUPERSEDED

NOT_ALLOWED_IN_PRODUCTION

4. Canonical A10 sub-axes

Use the following official sub-axis codes:

- A10A — Protein Folding & Chaperone System
- A10B — ER Stress & Unfolded Protein Response
- A10C — Autophagy–Lysosome Clearance
- A10D — Protein Aggregation & Proteotoxic Burden

The corrected Axis list has authority over later legacy narrative sections.

Modules should map primarily to parent Axis A10. Sub-axis codes may be used as:
mechanism tags;

evidence tags;

scoring refinements;
follow-up and explanation tags.
Sub-axis tags must not create duplicate Module records.

5. Official CELA approval

CELA is officially confirmed as:

Network Number: 120

Permanent Network Code: TSPI-NET-0120

Network Name: Cognitive–Emotional Loop Architecture Network

Network Layer: Living Biological Network

Canonical Primary Axis: A34

Canonical Primary Domain: D10 — Neurological Regulation

Mapping Status: APPROVED

The source Network Atlas identifies CELA as Network No.120 and describes its scope as thought loops, rumination, stress amplification, HPA activation, microbiome interaction, and chronic disease propagation. □

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The official relationship is:

A34 ≠ CELA

A34

= Parent Biological Axis

CELA

= Living Biological Network No.120

CELA.primary_axis_code

= A34

CELA must not replace or rename A34.

Canonical CELA record

```
{
  "network_code": "TSPI-NET-0120",
  "network_number": 120,
  "legacy_network_id": "N120",
  "network_name_en": "Cognitive–Emotional Loop Architecture Network",
  "network_name_th": "เครือข่ายสถาปัตยกรรมวงจรความคิดและอารมณ์",
  "primary_axis_code": "A34",
  "primary_domain_code": "D10",
  "mapping_status": "APPROVED",
  "mapping_authority": "TSPI Domain Expert"
}
```

Potential relationships with A1, A2, A17, A33, or A39 may be evaluated later as Secondary Axis relationships. They are not automatically approved by the confirmation of A34 and must not create duplicate evidence scoring.

6. Source-priority rule

The source Master contains corrected canonical material together with older embedded material.

The source-priority order must be:

Corrected 39-Axis list

Key Correction

Canonical numbered 9-Step list

Final Architecture

Numbered N1–N180 Network Atlas

Older detailed narratives

Historical per-Axis Module lists

Legacy infographics and headings

When two sections conflict, the higher-priority source must prevail.

Known legacy artifacts

Outdated Network count

An older infographic refers to 160 Networks. The complete Atlas and Final Architecture confirm 180 Networks.

Canonical count = 180 Living Biological Networks

Older A35 wording

Legacy:

Organ Resilience & Functional Reserve

Canonical:

A35 — Visceral Organ Functional Reserve

Older per-Axis Module lists

These lists may contain useful historical intent but must be labelled:

MIGRATION_REFERENCE_ONLY

They must not be loaded directly into Production.

7. The 9 Restoration Steps require a clean machine-readable Registry

The canonical numbered order is:

S1 — Detoxification

S2 — Microbiome Ecosystem

S3 — Immune–Inflammatory Regulation

S4 — Redox Homeostasis & Oxidative Balance

S5 — Enhance Metabolic Function & Mitochondria

S6 — Autophagy & Cellular Clearance

S7 — Genomic Stability & Proteostasis

S8 — Genomic Regulation Homeostasis

S9 — Prakati: Bioelectrical & Physiological Coherence

The source Master contains later narrative numbering drift in which Bioelectrical and Physiological Coherence is sometimes described as Step 8. This is an editorial inconsistency.

For the engine:

Bioelectrical & Physiological Coherence

belongs to S9 — Prakati

The canonical numbered list must become the Restoration Step Registry.

The Developer must not derive Step codes from narrative headings.

8. Status of the 180-Network layer

The latest Master resolves:

Network identities: RESOLVED

Network numbers 1–180: RESOLVED

Network names: RESOLVED

Source scope descriptions: RESOLVED

CELA as Network 120: RESOLVED

CELA Primary Axis A34: APPROVED

However, the Atlas is not yet a complete computational graph.

For most Networks, the following fields still require normalization and approval:

primary_axis_code

secondary_axis_codes[]

upstream_network_codes[]

downstream_network_codes[]

edge_type

effect_sign

accepted_symptoms[]

accepted_markers[]

minimum_evidence

evidence_grade

Therefore, the correct status is:

Network identities and source scopes are resolved. The normalized Network-to-Axis

Crosswalk and the Network Edge Registry remain incomplete, except for specific mappings

explicitly approved by the Domain Expert, including CELA → A34.

9. Blocker 1 — Module Registry remapping

What the Developer is asking

The Developer has created a legacy-to-current Axis converter that:

applies apparently safe mappings;

quarantines ambiguous mappings;

waits for expert approval before clinical use.

This is the correct technical strategy. However, no Module should become clinically eligible merely because a conversion was automatically applied.

Official answer

Please submit the complete candidate Module remap for row-by-row or batch-by-batch expert review. The semantic-remapping process is approved. The individual Module mappings are not automatically approved.

Only rows with:

mapping_status = APPROVED

may enter clinical Module ranking or patient-facing reports.

Safe auto-conversion criteria

A conversion may be classified as technically safe only when:

it is a one-to-one semantic equivalent;

the biological meaning is unchanged;

no Axis split or merge is involved;

the Axis is not deprecated;

the mapping does not conflict with the Module's documented mechanisms or research.

Meaning-changed, renamed, split, merged, or ambiguous mappings must remain quarantined.

Required candidate-remap fields

module_code

canonical_module_name
phytocore_code
legacy_axis_number
legacy_axis_name
proposed_primary_axes[]
proposed_secondary_axes[]
conversion_type
conversion_confidence
mapping_rationale
supporting_evidence[]
mapping_status
reviewer_comment
axis_master_version

The Project Status correctly identifies this as the highest-impact current blocker. □

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10. Blocker 2a — Primary and Secondary Axes of Networks

Official rule

Each Network must have:

exactly one Primary Axis

zero or more Secondary Axes

The Primary Axis represents:

the dominant biological function of the Network;

its principal scoring anchor;

the Axis under which its direct evidence is primarily interpreted.

Secondary Axes represent:

clinically meaningful cross-Axis influences;

upstream or downstream biological effects;

relationships that do not define the Network's core identity.

Secondary Axis mappings must not cause one evidence item to be counted repeatedly.

Domain derivation rule

The Project Status says that the Domain is automatically derived from the confirmed Axis.

This should be expanded to:

Primary Domain

= derived from Primary Axis

Related Domains

= derived from Secondary Axes

Do not manually enter duplicate canonical Domain values that may contradict the Axis Registry.

Import rule

The Atlas columns named "Domain" and "Biological Axis" must initially be preserved as:

source_domain_label

source_biological_axis_descriptor

They must not automatically become canonical codes.

Some values in those source columns are:

functional labels;

sub-axis concepts;

Network categories;

mechanistic descriptions.

They are not always exact 12-Domain or A1–A39 names.

No string matching, nearest-name matching, or LLM semantic matching may become authoritative without expert approval.

11. Blocker 2c — Network-to-Network edges

The Developer correctly states that without Network edges, the AI cannot reliably identify Root Drivers or explain biological propagation.

The Developer should prepare a Candidate Network Edge Registry.

Required schema

edge_id

source_network_code

target_network_code

edge_type

direction

effect_sign

clinical_context

temporal_pattern

estimated_time_lag

supporting_evidence[]

evidence_status

evidence_grade

confidence

source_reference

mapping_status

reviewed_by

approved_by

version

Required edge types

CAUSAL

REGULATORY

ACTIVATING

INHIBITORY

COMPENSATORY

POSITIVE_FEEDBACK

NEGATIVE_FEEDBACK

ASSOCIATIVE

CONTEXT_DEPENDENT

HYPOTHETICAL

Only approved causal, regulatory, activating, inhibitory, or compensatory relationships may affect:

Root Driver scoring;

propagation scoring;

Network Harmonization priorities.

Associative or hypothetical edges may be displayed in the Differential Network Model but must not be written as confirmed causality and must not increase clinical scores.

12. Additional missing dependency: Module-to-Network Claim Registry

The Project Status assumes that confirming the Network-to-Axis Crosswalk will activate the 25% Network Match component.

That alone is not sufficient.

A valid Network Match requires two separate relationships:

Patient Active Network

↔ Module Supported Network

If the Module's Network relevance is inferred only because both the Module and Network share the same Axis, then Network Match merely duplicates Axis Match.

The engine therefore requires an approved claim-level registry:

module_code

claim_id

network_code

mechanism

claim_scope

evidence_status

evidence_grade

source_reference

review_status

The canonical rule should be:

Module eligibility

= approved Module-to-Axis mapping

Network Match

= approved Module-to-Network claim evidence

Networks refine ranking and biological precision. They must not replace or override Module-to-Axis eligibility.

Until this Registry exists:

network_match_status = NOT_ASSESSED

network_match_score = null

13. Clinical reference content

The Developer is waiting for three critical clinical datasets:

Laboratory Cut-off Registry

Per-claim Evidence Grade Registry

Per-Axis Minimum-Evidence Registry

These datasets must be versioned and clinically approved. They must not be generated from general knowledge by the LLM.

13.1 Laboratory Cut-off Registry

Required fields:

marker_code

unit

sex

age_range

pregnancy_status

clinical_context

direction_rule

optimal_range

subclinical_range

functional_range

pathological_range

critical_low
critical_high
reference_source
approval_status
version

If no approved threshold exists:

severity_status = NOT_CLASSIFIABLE
reason = NO_APPROVED_THRESHOLD

13.2 Claim-level Evidence Grade Registry

Evidence Grade must be assigned to:

Module

- + Claim
- + Axis
- + Network

It must not be assigned once to the entire product.

A Module may have different grades for different claims.

13.3 Per-Axis Minimum-Evidence Registry

Each Axis requires its own criteria:

core_evidence[]
supporting_evidence[]
optional_evidence[]
accepted_combinations[]
conflicting_evidence[]
minimum_confidence
minimum_completeness
assessment_state_allowed

There must not be one universal rule such as “two markers are enough” for all 39 Axes.

14. Final decision: severity and incomplete evidence

This decision now supersedes any earlier ambiguity.

Data Completeness must not numerically reduce the severity of an observed abnormality.

Severity represents the biological magnitude supported by the available evidence.

Confidence and Completeness represent the reliability and coverage of that assessment.

A severe measured abnormality remains severe even when the overall patient dataset is incomplete.

The following calculation is prohibited:

Final Severity = Severity × Data Completeness

Example of an invalid calculation:

Severity 90 × Completeness 0.30 = 27

That would incorrectly convert a severe abnormality into a mild finding.

Axis assessment states

NOT_ASSESSED
AXIS_CANDIDATE
PROVISIONALLY_ASSESSED
ASSESSED
HIGH_CONFIDENCE_ASSESSED

NSS inclusion rules

NOT_ASSESSED

→ excluded from NSS

AXIS_CANDIDATE

→ excluded from final NSS

PROVISIONALLY_ASSESSED

→ may enter only a labelled Provisional NSS

ASSESSED

→ may enter NSS

HIGH_CONFIDENCE_ASSESSED

→ may enter NSS

Final NSS structure

The final clinical NSS should be calculated only from Axes that satisfy the required assessment state.

Data Completeness and Confidence should not act as severity multipliers. They should be reported separately.

Every NSS output must show:

Observed NSS

Assessment Coverage

Overall Confidence

Biological Uncertainty

Assessment Status

NSS Algorithm Version

Example:

Observed NSS: 68/100

Assessment Coverage: 42%

Overall Confidence: Moderate

Biological Uncertainty: High

Assessment Status: Provisional

Algorithm Version: TSPI-NSS-v0.2

Critical alerts remain separate and always override NSS.

15. Review of completed refinements

Ferritin mapping

The revised treatment of ferritin is approved:

Primary context:

A26 — Stem Cell & Hematopoiesis

Secondary inflammatory relevance:

A1 — only when corroborating inflammatory evidence exists

Ferritin must use a CONTEXT_DEPENDENT direction rule because it is both an iron-storage marker and an acute-phase reactant.

Ferritin alone must not prove:

inflammation;

iron deficiency;
liver dysfunction;
gut malabsorption.

HOMA-IR

The engine correctly refuses to calculate HOMA-IR when units are unknown.

The derived record should retain:

input_values
input_units
formula_id
formula_version
calculated_value
calculated_at

Network identifiers

The three-identifier model is approved:

Permanent Code

Display Number

Legacy N-ID

Example:

Permanent Code: TSPI-NET-0120

Display Number: 120

Legacy ID: N120

The Permanent Code must never change when Networks are reordered.

Ungraded Modules

Approved:

NOT_YET_GRADED

Ungraded does not mean Grade D.

Safety categories

Approved:

PASS

PASS_WITH_MONITORING

HOLD

CONTRAINDICATED

INSUFFICIENT_SAFETY_DATA

Safety must never be reduced to a numeric score alone.

Unavailable Network Match

Approved:

NOT_ASSESSED

It must not become zero and must not be silently reweighted.

16. Safety should operate both before and after ranking

The full pipeline currently lists Module Match before Safety.

For the ideal TSPI architecture, use two safety stages:

Module Candidate Generation

→ Preliminary Safety Eligibility Gate

→ Module Ranking

→ Combination Safety and Interaction Review

→ Physician Approval

The preliminary gate removes:
absolute contraindications;
major interactions;
missing essential safety data;
pregnancy or organ-function exclusions.

The final combination review evaluates:
ingredient duplication;
mechanism duplication;
pill burden;
additive bleeding risk;
additive glucose-lowering effects;
hepatic and renal burden;
total regimen complexity.

A contraindicated Module should never receive a clinically visible ranking.

17. Correct role of the 39 Axes and 180 Networks

The phrase in the Project Status should be refined to:

The 39 Biological Axes are the canonical assessment and Module-eligibility layer. The 180 Living Biological Networks are the executable mechanistic reasoning layer. Networks validate biological coherence, construct differential biological models, represent propagation, identify candidate Root Drivers and compensatory responses, and refine prioritisation. Networks do not override approved Module-to-Axis mappings.

This preserves both:

- the stability of the 39-Axis architecture;
- the central network-based philosophy of TSPI.

18. Persistent Patient Biological Model

The immutable analysis record per run is approved, but TSPI also requires a persistent, versioned Patient Biological Model.

Analysis Run 1 — immutable

Analysis Run 2 — immutable

Analysis Run 3 — immutable

These feed:

Patient Biological Model v1

→ Patient Biological Model v2

→ Patient Biological Model v3

New evidence may:

CONFIRM

REFINE

REDIRECT

OVERRIDE

CONTRADICT

a prior model.

The engine must preserve all previous versions and the reason for every model transition.

The AI must learn the behaviour of Networks over time, not merely memorise disease labels or automatically alter global Axis weights.

19. Report requirements

Physician Report

Must include:

source data and provenance;
red flags and critical alerts;
confirmed findings;
probable findings;
unresolved findings;
standard medical differential;
Primary and Secondary Axes;
Differential Network Model;
candidate Root Drivers;
compensatory Networks;
supporting and contradicting evidence;
missing evidence;
Module eligibility and ranking;
safety status;
reason selected;
reason not selected;
physician edits and approval.

Patient Report

Must clearly separate:

Confirmed

Probable

Not Yet Known

Suggested Next Tests

Clinician-Approved Plan

Safety Warnings

The Patient Report must not contain:

invented Root Causes;

unapproved Modules;

hypothetical values presented as patient findings;

simulated Omics values;

unvalidated outcome predictions;

AI-only prescriptions.

Multi-Omics Report

Generate a full Multi-Omics Report only when actual Omics data exist.

When Omics data are absent, generate:

Omics Data Readiness Report

Do not generate projected patient-specific values for:

cytokines;

bacterial abundance;

biological age;

metabolites;

proteomics;

pharmacogenomics.

20. Statement-level provenance

Every clinically meaningful statement must be traceable to its evidence and rule.

Required structure:

statement_id

statement_text

source_data_ids[]

rule_id

axis_codes[]

network_codes[]

module_codes[]

evidence_status

evidence_grade

confidence

data_completeness

biological_uncertainty

registry_versions[]

clinician_decision

A hypothesis must have:

scoring_effect = 0

It may trigger:

a follow-up question;

a suggested test;

inclusion in the Differential Network Model.

It must not raise Axis, Network, NSS, or Module Match scores.

21. Clinical validation and pilot requirements

The 71 automated tests must be supplemented with clinician-authored gold-standard cases.

Required test scenarios include:

symptoms only;

incomplete laboratory data;

conflicting evidence;

critical result with low NSS;

active red flag;

no Omics;

pregnancy;

renal or hepatic impairment;

contraindicated Module;

duplicate mechanisms;

legacy Axis conflict;

thalassemia versus iron deficiency;

normal imaging with persistent functional symptoms;

unapproved Network mapping.

The first physician pilot must be:

Clinician-facing

Shadow Mode

No automatic prescribing

No automatic patient release

The pilot must measure:

Axis agreement with clinicians

Network candidate agreement

Red-flag detection

Contraindication detection

Unsupported-statement rate

Clinician override rate

Cross-report consistency
Time saved
Adverse events
Patient comprehension
22. Updated project status
Software Core:
SUBSTANTIALLY COMPLETE

Technical Automated Tests:
PASSING FOR CURRENT IMPLEMENTATION

Canonical 3 Keys:
RESOLVED

Canonical 9-Step Numbered Registry:
REQUIRES FINAL NORMALISATION

Canonical 12 Domains and 39 Axes:
RESOLVED

A10 Sub-Axes:
RESOLVED

180 Network Identities, Names and Source Scopes:
RESOLVED

CELA as Network 120:
RESOLVED

CELA Primary Axis A34:
APPROVED

Network-to-Axis Crosswalk:
PARTIALLY APPROVED / REMAINDER PENDING

Network Edge Registry:
PENDING

Module Registry Remap:
PENDING EXPERT APPROVAL

Module-to-Network Claim Registry:
REQUIRED / PENDING

Clinical Reference Registries:
PARTIALLY COMPLETE

NSS Technical Implementation:

IMPLEMENTED

NSS Clinical Validation:
NOT COMPLETE

Module Match Technical Implementation:
IMPLEMENTED

Module Match Clinical Validation:
NOT COMPLETE

Physician Shadow Pilot:
NOT YET APPROVED

Patient-Facing Release:
NOT PERMITTED

23. Final Developer Lock

The following decisions are now final:

The corrected 39-Axis Master is authoritative.

A10 contains proteostasis, UPR, autophagy, lysosome function, and proteotoxic burden.

A26 is Stem Cell & Hematopoiesis.

A27 is Glycation / Carbonyl Stress.

A34 is Neuro–Sleep–Stress–Brain Clearance.

A37–A38 are the Oncology Axes.

A39 is Genomic Regulation Homeostasis.

CELA is Living Biological Network No.120.

CELA's canonical Primary Axis is A34.

CELA's canonical Primary Domain is D10.

Missing data must be NOT_ASSESSED.

Hypotheses never score.

Data Completeness does not numerically reduce observed severity.

Networks refine reasoning but do not override approved Module-to-Axis mappings.

Network Match requires an approved Module-to-Network Claim Registry.

Safety is a categorical exclusion and monitoring system.

Global learning is propose-only.

Reports require physician approval.

No simulated Omics values may be presented as patient findings.

Technical completion does not equal clinical validation.

Final conclusion

The development team is now working on the correct foundation. The software does not need to be rebuilt from zero.

The remaining work is to transform the canonical TSPI knowledge into a:
machine-readable;

semantically normalized;

clinically approved;

evidence-traceable;

version-controlled;

auditable;

fail-closed;

longitudinal Biological Intelligence System.

The fastest correct route is:

1. Approve the candidate Module remap.
2. Complete the N1–N180 Network-to-Axis Crosswalk.
3. Create and review the Network Edge Registry.
4. Create the Module-to-Network Claim Registry.
5. Approve the pilot Clinical Reference Registries.
6. Lock the final NSS and incomplete-evidence rules.
7. Run gold-standard clinical tests.
8. Conduct a physician shadow pilot.
9. Release patient reports only after formal approval.

This structure will allow TSPI to reason from Living Biological Networks, preserve the stable 39-Axis architecture, recommend only real approved Modules, and prevent the AI from guessing, hallucinating, inventing unsupported relationships, or converting missing evidence into false clinical certainty.

Additional TSPI Development Requirements

Clarifications to Prevent Semantic Errors, Duplicate Scoring, Unsupported Inference, and AI Hallucination

The current TSPI software foundation is strong. The Project Status confirms that the deterministic pipeline, evidence model, red-flag screening, phenotype engine, provisional NSS, Module Match, legacy-axis converter, versioned learning model, and Marker Registry have been technically implemented. It also confirms that Module approval, Network mappings, Network edges, and clinical reference content remain on the critical path. □

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The following requirements should now be added to the formal development specification.

They are intended to ensure that the system does not merely use correct names, but also uses correct relationships, correct evidence, correct scoring, and correct clinical governance.

1. Separate canonical ontology from patient-specific assessment

The system must strictly separate permanent TSPI knowledge from the assessment of an individual patient.

Canonical ontology

Canonical ontology defines what an entity is:

A34 = Neuro–Sleep–Stress–Brain Clearance

TSPI-NET-0120

= Cognitive–Emotional Loop Architecture Network

TSPI-NET-0120.primary_axis_code

= A34

Patient-specific assessment

Patient assessment determines whether that Axis or Network is active in a particular patient:

A34 assessment_status = PROVISIONALLY_ASSESSED

A34 severity = 62

CELA evidence_status = CLINICAL_INFERENCE

CELA confidence = MODERATE

The approval of:

CELA.primary_axis_code = A34

does not mean that CELA is automatically active in every patient with stress, insomnia, anxiety, rumination, or headache.

Required rule

Ontology defines what an Axis or Network is. Patient assessment determines whether it is biologically relevant or active in the current patient.

Ontology records and patient-assessment records must use different tables or schemas.

2. Do not use text similarity as authoritative mapping

The engine must not approve a mapping merely because words appear similar.

The following logic is prohibited:

Contains "inflammation"

→ automatically map to A1

Contains "stress"

→ automatically map to A34

Contains "gut"

→ automatically map to A17

A valid mapping must be based on:

canonical concept

+ official biological definition

+ documented mechanism

+ clinical context

+ approved crosswalk

String matching, embeddings, semantic similarity, or LLM suggestions may generate a candidate mapping, but they must never produce an authoritative mapping without expert approval.

Recommended statuses:

AUTO_SUGGESTED

CANDIDATE

CLINICALLY_REVIEWED

APPROVED

REJECTED

3. Implement evidence de-duplication

A single evidence item may support several interpretations, but it must not be counted at full numerical strength several times.

For example, one ferritin result may be relevant to:

iron storage;

hematopoiesis;

inflammatory activity;

liver context;

metabolic context.

However, the same ferritin result must not independently contribute its full weight to A26, A1, A15, and A35 unless an approved rule defines genuinely independent scoring components.

Required structure

evidence_item_id

primary_scoring_target

secondary_supporting_targets[]

scoring_allocation

duplicate_scoring_block

independence_rule_id

Required rule

One patient evidence item may support multiple biological interpretations, but it may contribute its full numerical weight only once unless an approved Registry rule explicitly defines independent scoring contributions.

Secondary mappings may support interpretation and confidence without automatically increasing severity.

4. Preserve the difference between null, zero, false, and NOT_APPLICABLE

These values must never be treated as interchangeable.

null

= no result or assessment is currently available

0

= the variable was assessed and its numeric value is zero

false

= the evaluated condition is not present

NOT_APPLICABLE

= the rule does not apply to this patient or clinical context

NOT_ASSESSED

= the entity has not been sufficiently assessed

Example:

network_match_score = null

network_match_status = NOT_ASSESSED

means the Network Match component cannot yet be evaluated.

It is fundamentally different from:

network_match_score = 0

network_match_status = ASSESSED

which means the Module was evaluated and showed no Network compatibility.

This distinction must be enforced at:

database level;

API level;

scoring level;

user-interface level;

report-generation level.

5. Create a Module-to-Network Claim Registry

Module-to-Network Match must not be inferred merely because a Module and a Network share the same Axis.

The following logic is prohibited:

Module maps to A34

Network maps to A34

Therefore, the Module treats that Network

Sharing an Axis establishes broad biological eligibility, but it does not prove a specific Network-level mechanism.

Required separation

Module eligibility

= approved Module-to-Axis mapping

Network Match

= approved Module-to-Network claim evidence

Required Registry

claim_id

module_code

network_code

claim_name

mechanism

claim_scope

target_axis_codes[]

evidence_status

evidence_grade

source_references[]

limitations

review_status

reviewed_by

approved_by

version

Until this Registry exists:

network_match_status = NOT_ASSESSED

network_match_score = null

The 25% Network Match component must not be generated from Axis co-membership alone.

6. Implement two-stage safety processing

Safety should run both before ranking and after a multi-Module regimen is assembled.

Stage 1 — Module eligibility safety gate

Before ranking, check:

absolute contraindications;

allergies;

pregnancy and lactation;

age restrictions;

renal impairment;

hepatic impairment;

major drug–herb interactions;

essential safety data availability;

disease-specific exclusions.

If an absolute exclusion is present:

module_safety_status = CONTRAINDICATED

ranking_allowed = false

If essential safety information is missing:

module_safety_status = INSUFFICIENT_SAFETY_DATA

ranking_allowed = false

A contraindicated Module must not receive a clinically visible rank.

Stage 2 — Regimen-level safety review

After several Modules have been selected, the system must check:

duplicated ingredients;

duplicated mechanisms;

additive bleeding risk;

additive glucose-lowering effects;

additive blood-pressure effects;

combined liver burden;

combined renal burden;

overlapping bowel-regulating Modules;

pill burden;

schedule complexity;

adherence feasibility.

A Module may be safe alone but inappropriate in a specific combination.

Recommended workflow:

Module Candidate Generation

→ Preliminary Safety Gate

→ Module Ranking

→ Regimen Composition

→ Combination Safety Review

→ Physician Approval

7. Define a complete score contract for every score

No score should appear without an auditable definition.

Every score must state:

what it represents;

how it was calculated;

which version of the formula was used;

which patient inputs were used;

which components were unavailable;

assessment coverage;

confidence;

biological uncertainty;

whether the result is provisional or final.

Required schema

score_id

score_name

score_value

score_status

formula_id
formula_version
input_evidence_ids[]
missing_components[]
assessment_coverage
confidence
data_completeness
biological_uncertainty
calculated_at
clinical_validation_status

The following type of output is not acceptable:

Module Match = 82%

unless the system also shows what the 82% means and how it was produced.

8. Root Driver must be confidence-graded

The AI must not present an inferred Root Driver as an unquestionable fact.

Use structured states:

CONFIRMED_ROOT_DRIVER
PROBABLE_ROOT_DRIVER
POSSIBLE_ROOT_DRIVER
COMPENSATORY_NETWORK
DOWNSTREAM_NETWORK
PROTECTIVE_NETWORK
NOT_YET_ASSESSABLE

The system should say:

This is the best-supported current biological model based on the available evidence.

It should not say:

This is the true root cause.

unless a sufficiently specific and clinically approved evidence standard has been met.

Every Root Driver conclusion should include:

supporting_evidence[]
contradicting_evidence[]
missing_discriminating_evidence[]
confidence
biological_uncertainty
alternative_network_models[]

9. Network edges must include context and time

A relationship between two Networks is rarely universal.

The influence of Network A on Network B may depend on:

acute versus chronic state;

infection status;

fasting or fed state;

age;

sex;

tissue;

pregnancy;

medication;

cancer stage;

disease stage;

exposure duration;
compensatory capacity.
Required edge fields
edge_id
source_network_code
target_network_code
edge_type
effect_sign
clinical_context
population_context
tissue_context
temporal_pattern
estimated_time_lag
reversibility
dose_or_exposure_dependency
evidence_status
evidence_grade
confidence
mapping_status

An edge must not be treated as universally active merely because it exists in the Registry.

10. Maintain immutable analysis runs and a separate persistent patient model

The system should maintain two different longitudinal structures.

Immutable analysis run

Each analysis event is permanently preserved:

Analysis Run 1

Analysis Run 2

Analysis Run 3

An old run must not change when:

a Registry is updated;

a mapping is corrected;

an algorithm changes;

new patient information arrives.

Persistent Patient Biological Model

The patient's current biological interpretation evolves:

Patient Biological Model v1

→ v2

→ v3

Every transition must specify why the model changed:

NEW_EVIDENCE

CLINICIAN_OVERRIDE

REGISTRY_UPDATE

DATA_CORRECTION

OUTCOME_RESPONSE

CONTRADICTING_FINDING

ALGORITHM_UPDATE

The previous model versions must remain available for audit.

The system should support:

CONFIRM
REFINE
REDIRECT
OVERRIDE
CONTRADICT

as explicit relationships between model versions.

11. Build a clinician-authored clinical regression suite

The 71 automated tests are useful technical tests, but they must be supplemented by clinical regression cases created and approved by clinicians.

The test set should include:

- symptoms only;
- incomplete laboratory data;
- contradictory laboratory results;
- a severe measured abnormality with low overall data completeness;
- a critical laboratory result with a low NSS;
- an active red flag;
- no Omics data;
- pregnancy;
- chronic kidney disease;
- liver impairment;
- an absolute Module contraindication;
- duplicated Module mechanisms;
- a legacy Axis conflict;
- thalassemia versus iron-deficiency differential;
- normal imaging with persistent functional symptoms;
- an unapproved Network mapping;
- a Registry version withdrawal;
- a clinically inappropriate but highly ranked Module.

Each case should define:

- expected_red_flag_action
- expected_axis_states
- expected_network_candidates
- expected_exclusions
- expected_module_eligibility
- expected_safety_status
- expected_report_language
- prohibited_conclusions[]

These cases must be rerun whenever:

- code changes;
- Registry content changes;
- a scoring formula changes;
- a Report Template changes.

12. Add an unsupported-statement detector

Before a report can be released, every clinically meaningful statement must have:

- source data;
- an approved rule or Registry record;
- Evidence Status;
- provenance;

the correct uncertainty label.

Required validation

statement_id

source_data_ids[]

rule_id

registry_record_ids[]

evidence_status

confidence

authoritative

If a generated statement has no support, the system must either:

remove it; or

label it explicitly as:

HYPOTHESIS — NOT USED FOR SCORING

The LLM must not add plausible-sounding clinical details merely to make the report read more smoothly.

13. Validate cross-report consistency at the structured-data level

Patient, Physician, and Multi-Omics reports must come from the same immutable analysis record.

Consistency checks must compare structured data, not only generated wording.

The following must remain identical across reports:

Axis code;

Axis assessment status;

Axis severity;

Network identity;

Network status;

selected Modules;

safety decisions;

missing-data status;

red flags;

clinician-approved plan.

Reports may differ only in:

level of detail;

language;

technical vocabulary;

educational explanation.

The system must not independently reanalyse the case separately for each report type.

14. Record structured reasons for Module non-selection

Matching is not prescribing.

For every Module that was eligible or matched but not selected, record a structured reason:

CONTRAINDICATED

MAJOR_INTERACTION

INSUFFICIENT_SAFETY_DATA

DUPLICATE_INGREDIENT

DUPLICATE_MECHANISM

LOWER_EVIDENCE

LOWER_RANK_THAN_ALTERNATIVE

INSUFFICIENT_PATIENT_DATA

NOT_RESTORATION_STEP_COMPATIBLE

EXCESSIVE_MODULE_BURDEN
REGIMEN_COMPLEXITY
PHYSICIAN_REJECTED
DIAGNOSTIC_WORKUP_REQUIRED_FIRST

This allows the physician to understand whether the system:
forgot a Module;

excluded it for safety;
replaced it with a stronger alternative;
deferred it due to insufficient evidence.

15. Add explicit clinical release states

The system should not use only "Draft" and "Final."

Recommended release states:

INTERNAL_TEST
RESEARCH_ONLY
CLINICIAN_DRAFT
CLINICIAN_REVIEW_REQUIRED
APPROVED_FOR_SHADOW_PILOT
APPROVED_FOR_CLINICIAN_USE
APPROVED_FOR_PATIENT_RELEASE
REJECTED
WITHDRAWN

Any report generated from a PROVISIONAL, UNAPPROVED, or missing Registry must not reach:

APPROVED_FOR_PATIENT_RELEASE

The release engine should fail closed.

16. Create a Registry dependency graph

Every report must be linked to every Registry and algorithm version used to generate it.

Example:

Framework Master v3

Axis Master v3

Network Master v1

Network Crosswalk v2

Module Registry v7

Marker Registry v4

Cut-off Registry v2

Safety Registry v5

NSS v0.2

Module Match v1.0

Report Template v3

Required dependency record

analysis_run_id

registry_name

registry_version

record_ids_used[]

checksum

effective_date

approval_status

If a Registry is withdrawn or corrected, the system must be able to identify:

all affected analyses;
all affected reports;
all patients requiring review;
whether reanalysis is required.

17. Implement rollback, withdrawal, and clinical incident management

Before a physician pilot begins, the system must support clinical incident management.

Examples of incidents:

incorrect Module-to-Axis mapping;
incorrect critical threshold unit;
incorrect Network crosswalk;
unsupported report statement;
missed contraindication;
wrong Registry version;
duplicated evidence scoring.

Required incident record

incident_id
incident_type
affected_analysis_runs[]
affected_reports[]
affected_registry
affected_registry_version
clinical_risk_level
containment_action
rollback_version
patient_notification_required
reviewed_by
resolution
closed_at

The system must be able to:

deactivate a Registry version;
prevent new reports from using it;
identify existing affected reports;
restore the last approved version;
preserve the audit trail.

18. Monitor technical performance separately from clinical performance

Technical success and clinical success are not the same.

Technical monitoring

Track:

processing_time
API_error_rate
Registry_lookup_failure
schema_validation_failure
report_generation_failure
database_integrity_error

Clinical monitoring

Track:

red_flag_miss_rate
critical_alert_miss_rate

contraindication_miss_rate
unsupported_statement_rate
Axis_agreement_with_clinician
Network_agreement_with_clinician
Module_usefulness_rating
clinician_override_rate
cross_report_inconsistency_rate
adverse_event_rate

A system may be technically fast and reliable while still being clinically wrong.

19. Display uncertainty dimensions separately in the clinician interface

The clinician should be able to see:

Severity
Confidence
Data Completeness
Assessment Coverage
Biological Uncertainty
Evidence Status
Evidence Grade

These dimensions must not be collapsed into one color or one score.

For example:

Severity:
color scale

Confidence:
bar or star indicator

Data Completeness:
percentage

Biological Uncertainty:
Low / Moderate / High / Very High

Evidence Status:
Measured / Derived / Inference / Hypothesis

An Axis may have:

Severity = Severe
Confidence = Low
Completeness = 30%

The interface must communicate all three facts simultaneously.

20. Capture structured clinician overrides

The physician interface should support:

APPROVE
EDIT
REJECT
DEFER
MARK_UNSUPPORTED
REQUEST_MORE_DATA
CHANGE_PRIMARY_AXIS

ADD_SECONDARY_AXIS
REMOVE_SECONDARY_AXIS
SELECT_MODULE
REJECT_MODULE
OVERRIDE_DOSE
OVERRIDE_SAFETY_WITH_JUSTIFICATION

Every override must record:

original_output
final_clinician_decision
override_type
override_reason_code
free_text_rationale
reviewer_id
timestamp
eligible_for_learning

Possible reason codes include:

NEW_CLINICAL_INFORMATION
PATIENT_PREFERENCE
SAFETY_CONCERN
REGISTRY_ERROR
ALGORITHM_ERROR
CLINICAL_JUDGMENT
DIAGNOSTIC_UNCERTAINTY
TREATMENT_RESPONSE

The learning system must not treat every override as an algorithm error.

21. Official CELA confirmation

The following mapping is now formally approved:

network_code = TSPI-NET-0120

network_number = 120

network_name_en
= Cognitive–Emotional Loop Architecture Network

primary_axis_code = A34

primary_domain_code = D10

mapping_status = APPROVED

This means that CELA's canonical biological location is A34 — Neuro–Sleep–Stress–Brain Clearance.

It does not mean:

CELA is an Axis;

CELA replaces A34;

CELA is automatically active in every patient;

every stress-related symptom confirms CELA dysregulation.

Patient-specific CELA activation must still pass the relevant symptom, phenotype, minimum-evidence, confidence, and clinician-review rules.

22. Final governing principle

The AI must not only avoid inventing names. It must also avoid inventing relationships.

Even when every Axis, Network, and Module name comes from a real Registry, the report can still be wrong if the system:

- maps a marker to the wrong Axis;
- maps a Network to an unsupported Module;
- counts one evidence item repeatedly;
- converts association into causation;
- treats missing data as zero;
- reduces severe findings through data incompleteness;
- invents a Network edge;
- generates unsupported explanatory text;
- silently uses an unapproved Registry.

The ideal TSPI system must therefore govern all of the following:

Identity

Mapping

Evidence

Scoring

Safety

Explanation

Versioning

Approval

Outcome

Learning

Final instruction to the development team

No unapproved mapping, missing Registry, hypothesis, LLM-generated relationship, duplicated evidence contribution, or unsupported statement may silently enter Axis scoring, Network reasoning, Module selection, or patient-facing reports.

The LLM should remain the explanation, translation, and presentation layer. Authoritative clinical conclusions must come only from:

approved patient data

+ approved Registries

+ approved deterministic rules

+ validated calculations

+ clinician approval

This is the required architecture for TSPI to operate clearly, consistently, safely, and without guessing or hallucination.